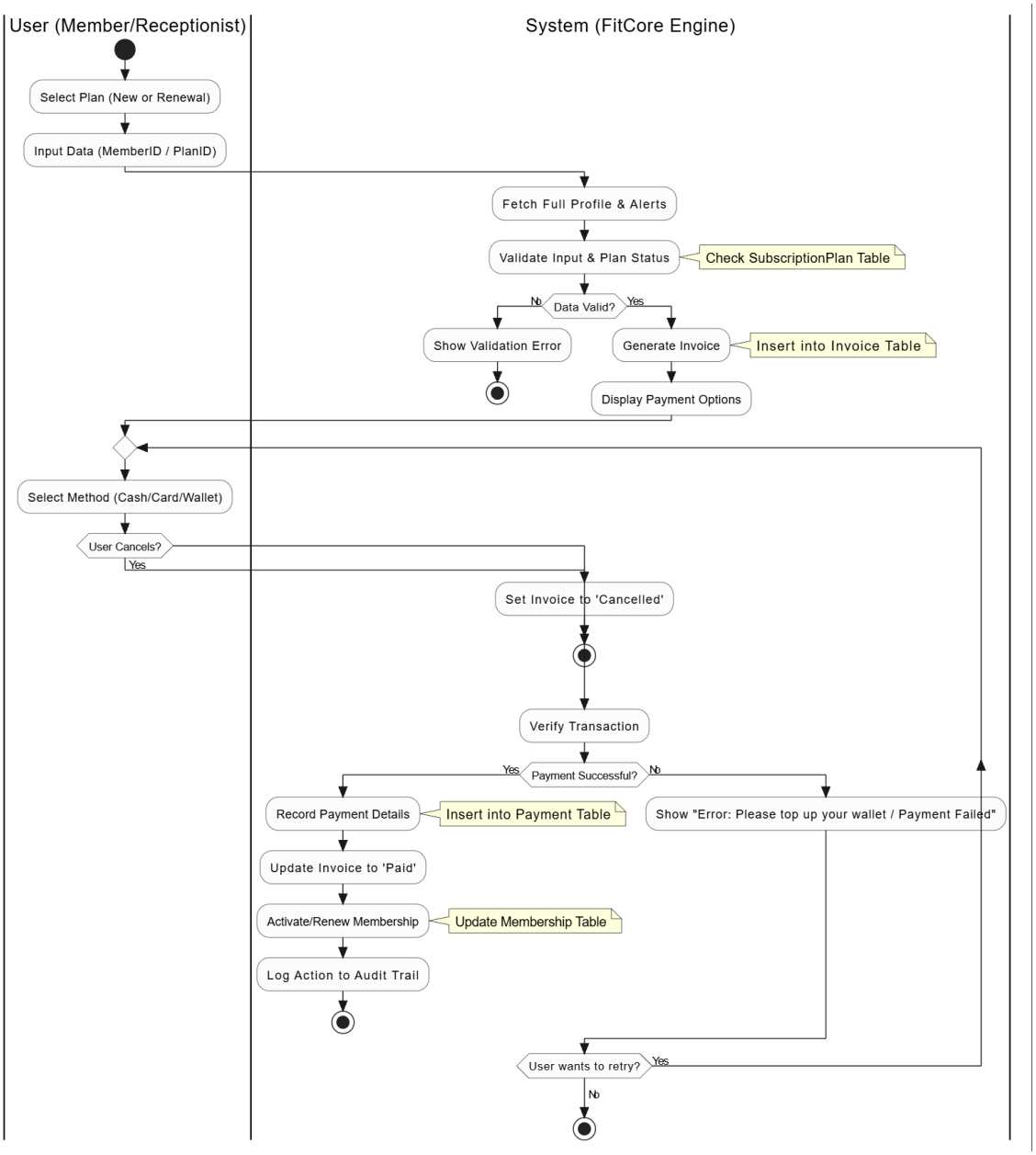
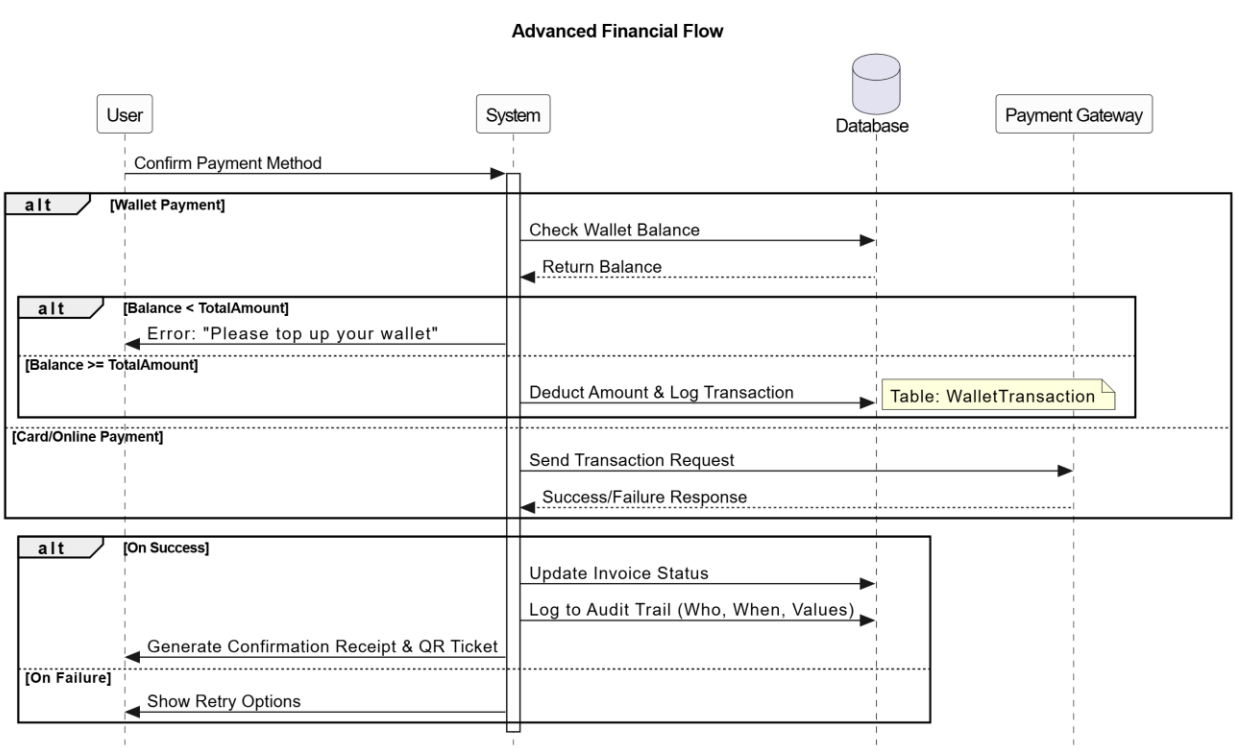
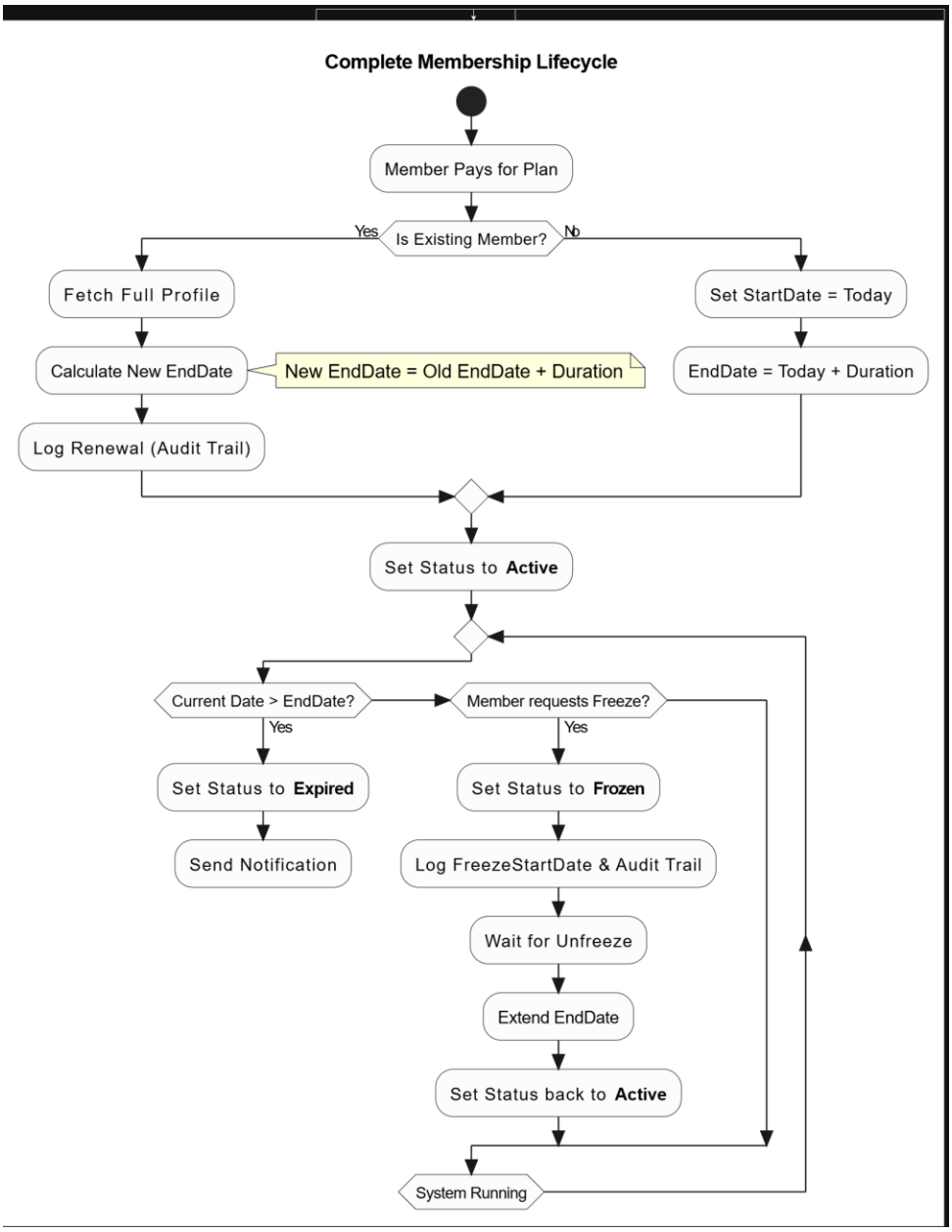
Activity diagram:




payment and invoices :




membership diagram :


Complete Membership Lifecycle
Member Pays for Plan
Is Existing Member?
Yes
No
Fetch Full Profile
Set StartDate = Today
Calculate New EndDate
New EndDate = Old EndDate + Duration
EndDate = Today + Duration
Log Renewal (Audit Trail)
Set Status to Active
Current Date > EndDate?
Member requests Freeze?
Yes
Yes
Set Status to Expired
Set Status to Frozen
Send Notification
Log FreezeStartDate & Audit Trail
Wait for Unfreeze
Extend EndDate
Set Status back to Active
System Running